Attitude Determination of Orbiting Objects from Lightcurve Measurements

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This paper describes a method based on virtual reality tools to achieve the attitude determination of an orbiting object using lightcurve measurements. A virtual model of the orbiting object is propagated in order to reproduce its lightcurve. The differences between the real and simulated lightcurves are used as the cost function to be minimized through a multiobjective genetic algorithm. Lightcurve measurements in different spectral bands, both with monostatic and multistatic optical observations can be used. The method is applied to the real measurements taken from Cerro Tololo and Flagstaff Observatory of the spacecraft GSAT3. A set of attitude parameters compatible with the measured lightcurves is identified and discussed.

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NOMENCLATURE

φ_0	Initial roll angle.
θ_0	Initial pitch angle.
ψ_0	Initial yaw angle.
p_0, q_0, r_0	Initial rotation velocity in body frame.
CCD	Charge-coupled device.
FFT	Fast Fourier transform.
GEO	Geostationary earth orbit.
LF	Light flux.
MLI	Multilayer insulation.
RGB	Red, green, blue.
$q1, q2, q3$	Quaternion vectorial components.
$q4$	Quaternion scalar component.

I. INTRODUCTION

The Recent studies in the field of space debris optical measurements address the analysis of space object lightcurves [1]–[3]. These data can be related to the space debris attitude dynamics and can be exploited to identify their motion. The problem of lightcurve inversion has been extensively studied for asteroids [4]–[6] not only for attitude determination but also to identify physical features (i.e., shape, materials reflectance etc.). In the case of space debris, a number of assumptions can be made on the shape and on the materials of the target; hence, the problem of lightcurve inversion is mainly related to the attitude motion or to the evaluation of optical properties of aged materials [7]–[10]. Methods based on radar [11] or lidar [12] can be exploited to analyze the space debris attitude dynamics in low earth orbit. In higher orbital regimes, only optical measurements can be used.

This past research has great importance from a theoretical point of view, to confirm the effects of the environmental torques on space debris and to improve the accuracy of complex models for the orbital determination of high area-to-mass ratio objects. The trajectories of these objects are greatly influenced by environmental disturbances related to their shape and attitude motion, induced by solar radiation pressure as well as gravitational torque [13], [14]. Moreover, these analyses have a practical impact in the definition of possible targets for active debris removal missions. In this case, the accurate identification of attitude motion of uncooperative objects is a focal point in the design of the mission. At present, joint optical observations methods have been studied to increase the accuracy of orbit determination [15]–[17], but these methods can also be used to improve knowledge on the attitude motion.

In this paper a method for attitude motion identification through lightcurve analysis based on a three-dimensional (3-D) virtual reality model of the spacecraft is proposed. The method is applied to the attitude determination of the spacecraft GSAT-3, using the lightcurve measurements of a joint observation campaign carried out from the USNO observatory (Flagstaff—USA) and the MODEST observatory (Cerro Tololo—Chile). This was chosen as a study case in virtue of the complex shape of the spacecraft [18].

II. OBSERVATION CAMPAIGN

The data reported here was obtained on two telescopes [2].

1) The University of Michigan's 0.6-m-aperture Curtis-Schmidt telescope, also known as MODEST, for Michigan Orbital DEbris Survey Telescope, located at Cerro Tololo Inter-American Observatory in Chile. This telescope is equipped with a single E2V thinned, backside illuminated CCD with 1.45 arcsec/pixel and a field of view of 1.6° × 1.6°.
2) The U.S. Naval Observatory Flagstaff Station's 1.3-m telescope located outside of Flagstaff, AZ,USA. This telescope has a 2 × 3 mosaic of thinned CCDs with a field of view of 1.02° × 1.36° and a scale of 1.2 arcsec/pixel.

Since this was the first attempt at coordinated observations between the two stations, the observing strategy was to observe a number of controlled and uncontrolled objects at GEO in as simple a manner as possible. Observing sequences for each object were 30 min long, starting on the hour or half-hour. Each telescope was positioned at the appropriate hour angle and declination such that the specified target would pass through the middle of the field of view at the halfway point of the sequence. Each site took data as rapidly as possible since it was not possible to synchronize the camera shutters at these widely separated sites.

The cadence was 19 s for MODEST and 38 s for the USNO 1.3 m.

MODEST observed through a broad V+R filter while USNO 1.3-m used a Sloan r filter.

Both sites observed the same Landolt (2009) standards for photometric calibration. All observations were referenced to the Landolt R magnitude.

Every image was visually inspected to check for contamination of the object aperture by star streaks. Any image with such contamination was not included in the final analysis. Magnitudes were measured through circular apertures, and corrected for extinction and zeropoint as determined by observations of standard stars. No corrections were made to a standard distance and phase angle.

The difference in the lines of sight from the two sites to the spacecraft is about 10°.

III. SATELLITE MOTION ANALYSIS THROUGH VIRTUAL REALITY MODEL

A. Attitude Motion Determination Tool Overview

A possible way to solve the attitude motion determination of satellite with complex shape from lightcurve is to use a 3-D virtual reality model of the satellite itself.

Virtual reality models give the possibility to reconstruct the real satellite in every particular. This model can be made very accurate, provided that all details of spacecraft external surfaces, including shape and light reflection properties are known. Nevertheless, even a raw 3-D virtual model of the satellite permits to achieve meaningful results, as discussed in the following sections of this paper.

In the virtual reality model used in this paper the satellite motion is based on the torque-free rigid body Euler equations of motion. More accurate dynamical models are possible, but in general not necessary for ligthtcurve simulation. The satellite attitude is determined by the minimization of the cost function represented by the root mean square of the residuals between measured and simulated lightcurve.

The three main parts of the system are as follows:

1) 3-D virtual world simulating the satellite and reproducing its lightcurve based on effective illumination and observation geometry (observer, satellite, Sun);
2) Attitude motion propagator implementing the torque-free rigid body rotation on the basis of initial conditions;
3) Minimization of residuals, obtained comparing the real and simulated lightcurve, through a global optimization tool based on genetic algorithms.

B. Discussion of the Virtual Model

The effectiveness of virtual reality tools is presented, implementing a 3-D virtual model of the satellite GSAT3. This satellite has a complex shape in terms of geometry, materials, and colors. The model was designed on the basis of the information found in the literature [18]. The main components identified as crucial for light reflection and sketched are as follows:

1) Body covered in MLI;
2) Three antennas (front white, rear MLI), curved shape;
3) Solar arrays (blue);
4) Two aluminum rectangles covering the body sides where solar arrays are attached (gray).

The front and rear view of the model is shown in Fig. 1.

The surface light reflection in the virtual model is based on the Phong lighting model [19], where ambient light intensity is set to 0 and diffuse Lambertian and specular components of light intensity are set in order to get the desired appearance of satellite's features.

The MLI has been imitated, according to [20], by assuming a coefficient of specular reflection of 0.6 and a coefficient of diffuse reflection of 0.3 both in red and green in order to obtain the yellow appearance (0 in blue). The solar array is dark blue, in accordance to their reflectance spectra [21]. Their specular coefficients are lower than 0.1, in virtue of their large absorptivity, and their contribution to lightcurve is related to their large surface more than reflectivity. The inner part of antennas was set to white with low diffuse coefficient and specular coefficient of 0.8 in all the channels (RGB). The aluminum of lateral panels was set to dark gray with low reflectivity in green and blue channels.

These choices come from the analysis of literature, based on GSAT3 published information and drawings.

The uncertainties on real reflectance values of different materials and their aging, as well as on the real dimensions of satellite parts and their apparent proportions during rotation is considered in the total error budget and taken into account in the optimization process during the comparison with the lightcurve measurements.

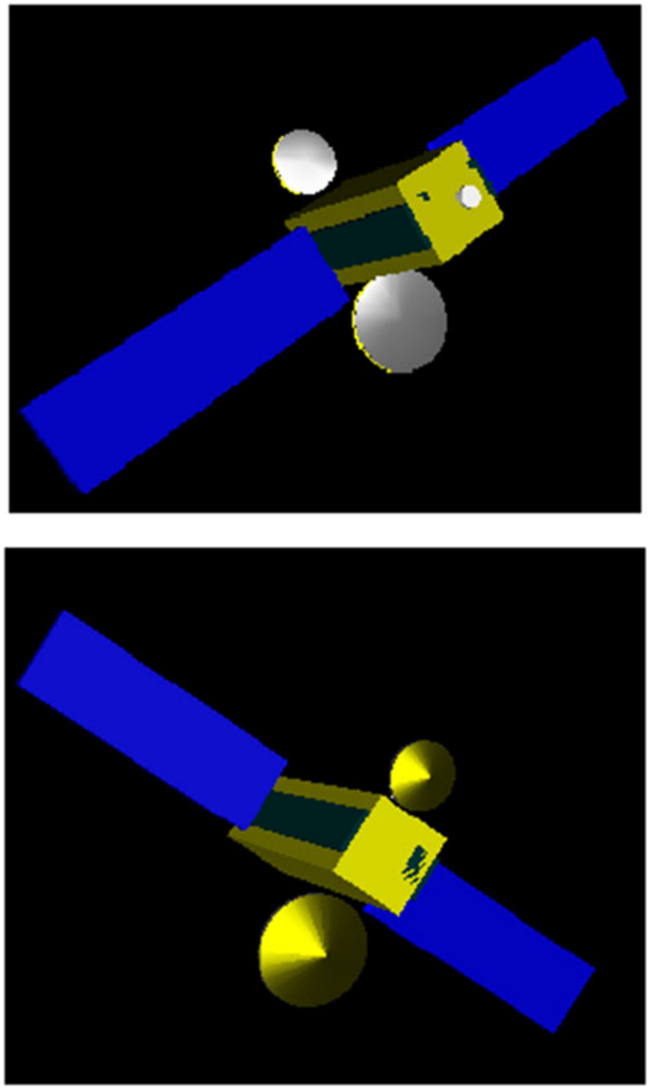

Fig. 1. GSAT3 virtual reality model showing front and rear views of antennas, solar arrays, and body.

Another source of errors is given by self-shadowing, which is not taken into account in the virtual model. This effect increases with the phase angle, and it is null at zero phase angle, when shadows are projected on surfaces that are not visible by the observer. In the GSAT3 study case the phase angle is 20°. Under this condition the solar arrays can project a shadow on the lateral surface of the satellite, which is maximum when the solar panel centerline, the observer, and the sun direction are coplanar, as sketched in Fig. 2. In this case, considering solar array five meters long and $\alpha = 10°$ (half of the phase angle), a projected shadow of about 86 cm is obtained on the lateral surface of the satellite. Considering a satellite length of 4 m, we obtain a worst case error of about 20% in light peaks, at the instant in which the solar array shadow is maximum, thus providing a reduction of the intensity of the reflection spike. The shadow length is proportional to $\sin\alpha$, thus its effect disappears when the phase angle tends to 0° (ideal configuration for observations in GEO).

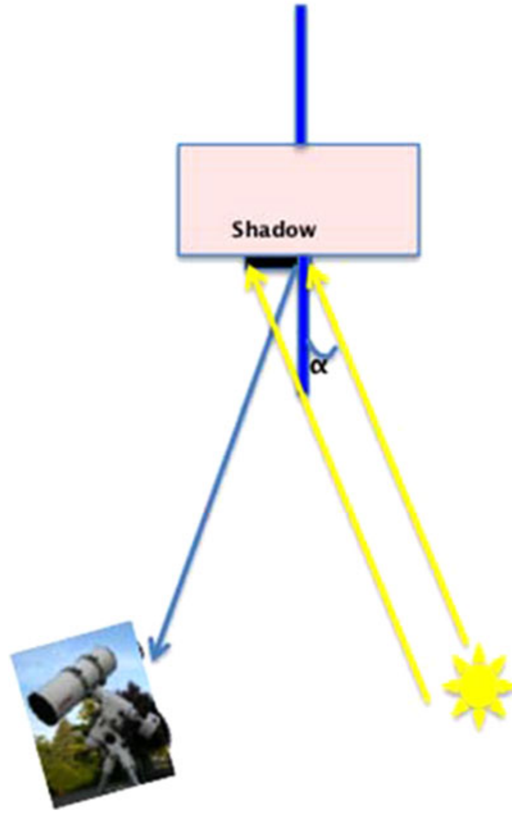

Fig. 2. Scheme of the observation geometry highlighting the presence of projected shadow.

Due to this consideration and since the important parameter in the following analysis is not the absolute intensity of lightcurve peaks, but only the presence itself of the peaks, the effect of self-shadowing was neglected.

C. Simulated Lightcurve: Sites and Colors

Satellites can be observed simultaneously from different sites and with different filters; both of these techniques offer the possibility to increase the satellite attitude observability, highlighting different particulars.

As an example, the lightcurves of GSAT3, obtained with a rotation rate of about 0.5°/s for every axis, have been simulated in RGB from Flagstaff and Cerro Tololo. The simulated lightcurves are sketched in Fig. 3.

We simulate in three passbands to show the power of observing in different filters simultaneously to determine spacecraft characteristics. The observations were taken only in the *R*-band, and so we limit our comparisons to that band.

The shape of the lightcurves from both sites is almost equal, showing sensible differences only in the peak heights due to different angular specular effect. The difference between lightcurves from the two sites increases when a larger number of particulars are included in the satellite model. Nevertheless, the differences among different color channels are well visible also from the same site even with a raw model of the satellite. In particular the reflectance effect on blue channel, due to solar arrays, is evident in Fig. 3.

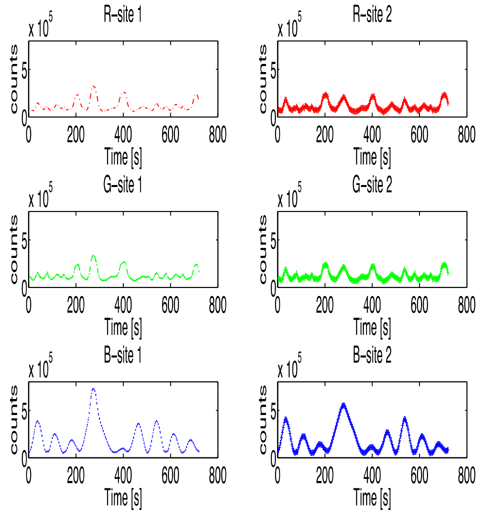

Fig. 3. Comparison of lightcurves of the virtual model in the red, green, and blue bands from two observation sites.

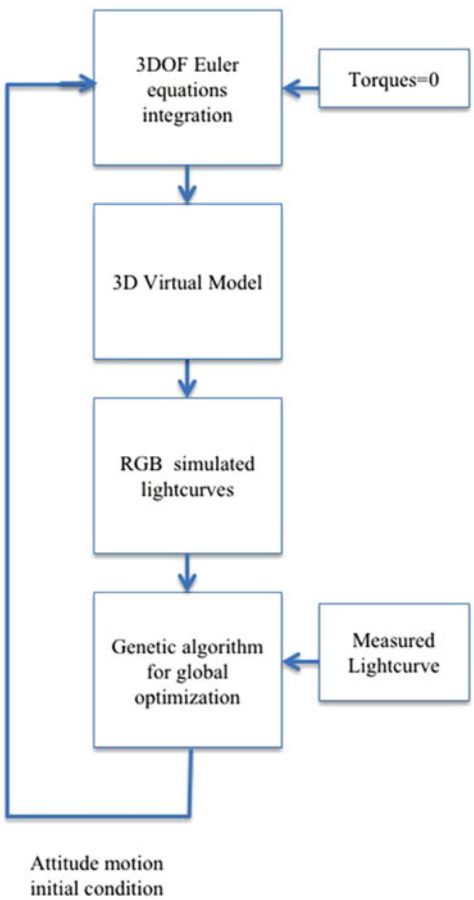

Fig. 4. Sketch of the algorithm for attitude determination.

D. Attitude Determination Tool

Once the 3-D virtual model has been set, it has to be processed in order to evaluate the lightcurve. The algorithm adopted is briefly sketched in Fig. 4.

The dynamic model is represented by the Euler equation of rigid body motion [22], the external environmental and control torques are not considered since the environmental torques effect is negligible during the period of acquisition of the lightcurve and the control torques do not exist for spent spacecraft or space debris. In the case study of GSAT, a diagonal inertia matrix with nondimensional moments of inertia (1, 2, 2) was considered, in order to represent the difference between the inertia around the solar arrays symmetry axis and the two transverse ones.

The inputs to the attitude motion propagation are the six initial conditions ($\varphi_0, \theta_0, \psi_0, p_0, q_0, r_0$) in terms of Euler angles and angular velocity components in the satellite reference frame.

These initial conditions are evaluated by the multiobjective global optimization tool based on genetic algorithms, in which cost functions J_m are given by the mean square roots of the differences between measured and simulated lightcurves from the two observation sites

$$J_m\left(\varphi_0, \theta_0, \psi_0, p_0, q_0, r_0\right) = \sqrt{\sum_n \left(\overline{\mathrm{LF}}_n - \mathrm{LF}_n\right)^2}$$

where $m = 1, 2$ represents the two observation sites, $n = 1, 2 \ldots$ is the nth measurement, and $\overline{\mathrm{LF}}$ and LF are the simulated and measured light flux, respectively.

The multiobjective genetic algorithm (MOGA) tool was chosen due to its capability to spread the research in a large dominium of the variables. The genetic algorithm is an optimization tool, which seeks to optimize the components of a vector-valued cost function [23]. Unlike single objective optimization, the solution to this problem is not a single point, but a series of points known as the Pareto-optimal set. If the observation sites produce measurements with different accuracy, it is possible to give solutions different weights. In this case, it should be possible to choose the Pareto solution that minimizes the cost function calculated with measurements from the most reliable site. In the present case, both sites have the same reliability, thus a central Pareto solution was chosen, optimizing at the same level both the components of the cost function.

The search for the optimum was limited to a bounded set, in order to avoid nonrealistic solutions. In particular, the angular rate components have been bounded to values lower than 20°/s. This choice avoids solutions including multiple turns between close measurements.

For the MOGA tools have been maintained the default MATLAB options which are available in [24].

It has to be noticed that the system works in the same way for different number of observation sites (one or multiple) just by increasing the number of objective cost functions and with lightcurves in different colors since the 3-D virtual model produces the simulated lightcurves in RGB channels.

The reliability of the proposed method is highly related to the shape and peculiarities of each target. The more the target has a complex shape, the higher the possibility to get the correct attitude motion. It is impossible, for instance, to reconstruct the motion of objects with spherically symmetrical surface. Under this consideration, before attempting to use the algorithm with real measurements, it is extremely

TABLE I
Simulated and Identified Initial Condition

Attitude parameter	Simulated parameter value	Optimization identified parameter
Pitch [rad]	0	3.1411
Roll [rad]	0	3.1408
Yaw [rad]	0	3.1376
p [rad/s]	0.01	0.0100
q [rad/s]	0	1e–5
r [rad/s]	0	3e–5

TABLE II
GSAT3 Identified Initial Conditions

Initial condition	Value
Pitch [rad]	3.9495
Roll [rad]	0.1881
Yaw [rad]	1.0916
p [rad/s]	0.00028
q [rad/s]	0.01454
r [rad/s]	0.01876

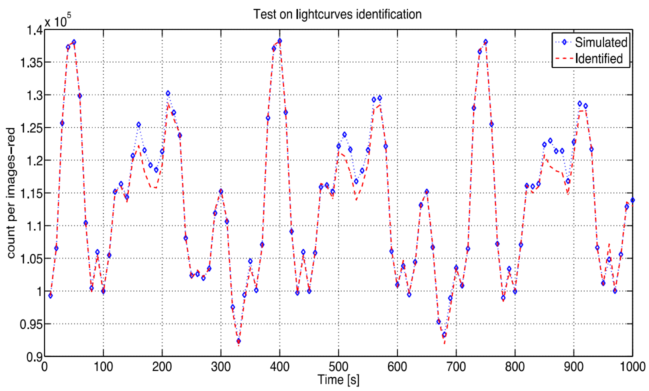

Fig. 5. Simulation of attitude motion identification, comparison between simulated, and identified lightcurve.

important to evaluate the degree of success of the attitude determination process for a particular spacecraft by using simulated noisy measurements. Only if the performance is satisfactory, it would be possible to apply this process to real cases.

A test of motion reconstruction by single site measurements was performed by evaluating the lightcurve with the 3-D virtual model of GSAT imposing the initial conditions and producing a measurement every 10 s for 1000 s. The measurements, in terms of pixel bit counts, were corrupted with zero-mean Gaussian noise with standard deviation equivalent to 0.1 magnitude. The optimization tool identified the initial conditions sketched in Table I.

The initial conditions identify the attitude and angular velocity correctly.

The comparison between lightcurve produced by simulated and identified attitude motion is given in Fig. 5. Some, minor discrepancies in peak intensity can be noticed at 180, 530, and 870 s, due to the noise on measurements producing an error on the initial state determination on the order of 0.01 rad (about 0.5°).

IV. GSAT3 SATELLITE ATTITUDE MOTION RECONSTRUCTION

A. GSAT3 Attitude Determination

The application of the presented method to real measurements requires a further passage to reduce simulated and real measures to comparable units of measure.

While the real measures are achieved in terms of magnitude, the simulated ones are given in terms of image pixel light intensity bit counts. Being interested in the relative appearance from sequent images given by the virtual model, it was decided to reduce the real measurements from magnitude to LF and from absolute LF to image counts. In this way, the optimization tool can compare two curves, ranging between zero and one.

Only the simulated curves in R-band were used in attitude determination algorithm and compared with real measurements since all observations were referenced to the Landolt R magnitude.

The resulting attitude parameters achieved by the GSAT3 attitude motion analysis are given in Table II.

The inverse procedure to represent the absolute values of lightcurve is not strictly needed for the procedure of attitude motion identification but it is useful to analyze the results.

In order to produce a lightcurve comparable, in magnitude, with the real one, a calibration procedure keeping into account the conversion factor from image counts to LF has been implemented.

The comparison between real and simulated calibrated lightcurve produced by the identified attitude motion are sketched in Figs. 6 and 7 for the two observation sites.

From the Flagstaff graph, we notice a good agreement between real and simulated lightcurves in the central part of the observations and some discrepancies at the beginning and at the end. In particular, at time 300 and 1400 s, we have a shape of the measured curve with small double peaks, reproduced by the simulated lightcurve only for the second peak, whereas in correspondence of the first peaks of the real curve the simulated one presents two valleys. A similar behavior is observed in some parts of the Cerro Tololo graph, in which the curves are very similar for large portion of the observation time, but in correspondence of 200, 700, and 1200 s a light peak in measurements is not reproduced in the simulated curve. These discrepancies, appearing in the data from both sites, could be due to reflective particulars of the satellite that could not be identified and be included in the virtual model, based on the data available in the literature. The available information allowed for the inclusion in the model only of the most relevant features, such as antenna, solar panel, and MLI covers locations, but it was not sufficient for a detailed reproduction of all the satellite external shape and reflective properties. Hence some discrepancies with respect to reality must be accepted, due to incorrect reproduction in the virtual model of details such as antenna alignment, antenna optical reflective properties, aging on MLI, and due to possible missing elements of

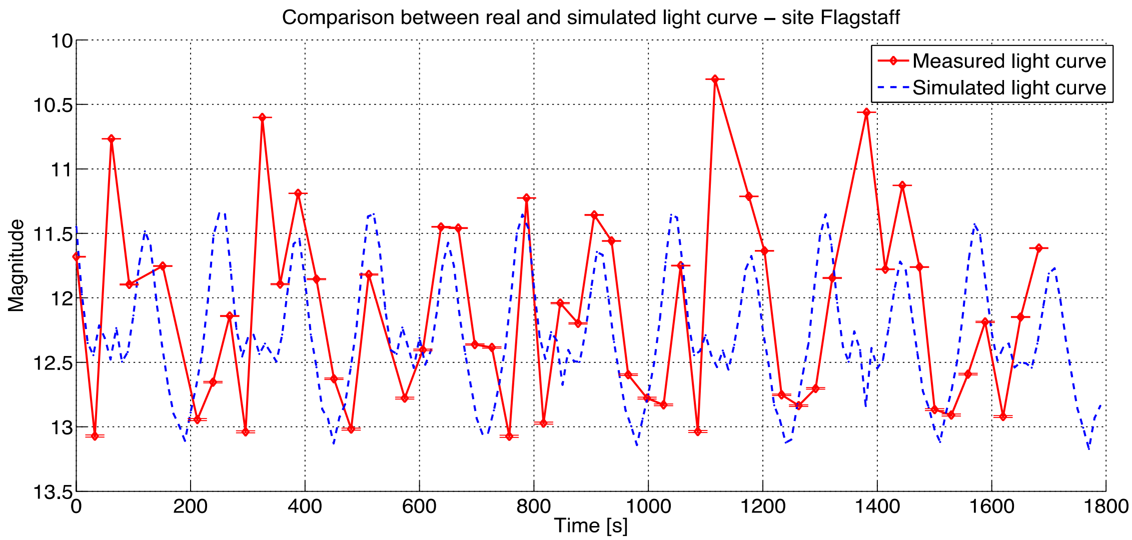

Fig. 6. Comparison between real measurements taken from USNO observatory in Flagstaff (red continue curve) and simulated lightcurve (blue dashed line) obtained by solving the attitude of GSAT3.

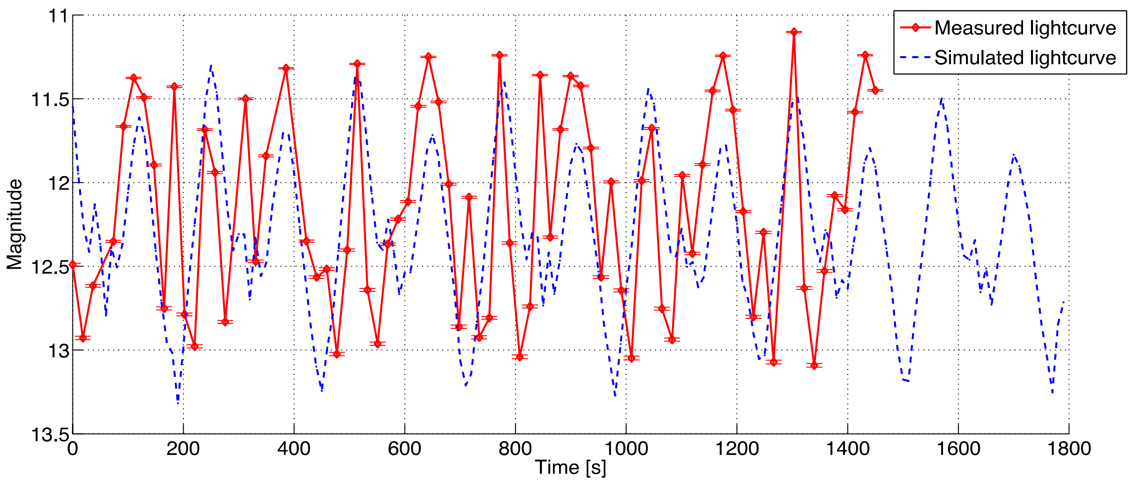

Fig. 7. Comparison between real measurements taken from MODEST observatory in Cerro Tololo (red continue curve) and simulated lightcurve (blue dashed line) obtained by solving the attitude of GSAT3.

the satellite, not shown in the satellite images and publicly available data.

In order to show an example of the effect of changes in the spacecraft geometry and surface reflection properties on the simulated lightcurve, antenna dimensions and reflective properties have been changed, obtaining the lightcurve shown in Fig. 8. This lightcurve, while capturing the global satellite behavior as the previous one, removes dark peaks in the Cerro Tololo reproduced curve appearing in Fig. 7.

By the comparison between the curves in Figs. 7 and 8, it is possible to observe that the sequence of peaks and valleys of the real and of the simulated lightcurves are still aligned, even if their relative values are different. This confirms that the relevant satellite properties originating in the lightcurve fluctuations are well captured by the virtual reality model, even if inaccuracies are present, due to less important particulars.

This discussion permits to evaluate the effective possibility to introduce the inverse problem, that is whether it is possible, once the attitude motion is determined, to identify discrepancies between virtual model and real spacecraft induced, as an example, by environment effects or malfunctions. Though not a trivial task, this seems feasible in principle. However, due to the large number of possible parameters that can affect the lightcurve, an accurate and reliable bidirectional light reflection model is necessary, based on bidirectional light reflection measurements while the satellite is on ground, before launch.

The present analysis is concentrated in reproducing in the simulated lightcurve the sequence of peaks and valleys of the measured lightcurve, showing the potentialities of the proposed approach. With the poor available data on the satellite shape and reflectivity, the goal cannot be the exact reproduction of the lightcurve magnitude. This is strictly related to having a reliable model of the spacecraft, which should be consistent with the spacecraft shape, materials, and possible aging effects.

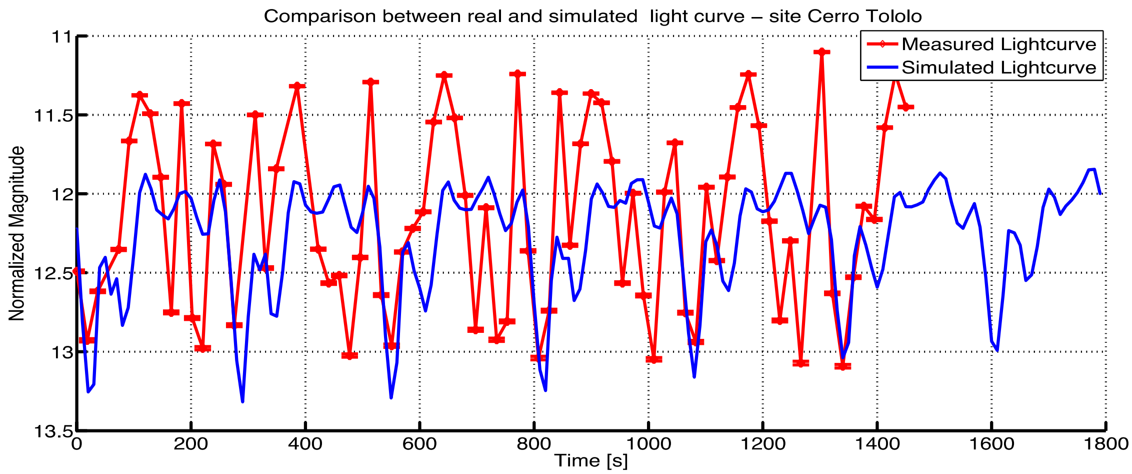

Fig. 8. Comparison between real measurements taken from MODEST observatory in Cerro Tololo (red continue curve) and simulated lightcurve (blue dashed line) obtained by solving the attitude of GSAT3 and manipulating GSAT3 model antennas dimensions and reflectivity.

As a result of the above discussion, it can be stated that a nondetailed spacecraft model can permit to reconstruct the attitude motion, which is very useful for space debris, but it cannot permit the evaluation of aging and environment effect on spacecraft physical properties.

It must be noticed that introducing changes in the basic spacecraft virtual model to be more adherent to one measurement curve, leads to larger differences with respect to the measurement curve taken from the other observatory. This highlights the importance of using multiple observatories with different lines of sight. As a matter of fact, combined observations permit not only to collect a larger number of data but also the possibility to include, in the optimization process, information about the timing in which some particulars are observed. For instance, light peaks or dark valleys observed only from one site imply that some luminous or dark particulars of the spacecraft are visible only from one line of sight, constraining the possible attitudes of the spacecraft at that time. Moreover, a multiple observation scheme permits to get data, almost simultaneously, in different color bands, thus increasing the information available to the optimization process, since some particulars of the spacecraft could be observed only in some colors.

To highlight the correlation between real and simulated lightcurves the Fourier analysis was applied to the curves from both sites. The results of main frequencies identified from FFT for both sites and for both real and simulated curves, are sketched in Fig. 9 and summarized in Table III.

The attitude determination tool permitted to identify some of the main frequencies present in the real lightcurves. In particular frequencies identical on both sites where identified by the tool and other frequencies highlighted only from one site.

The attitude determination tool evidences a lower number of frequencies than the real world because of the model simplifications, which do not take into account some satellite particulars.

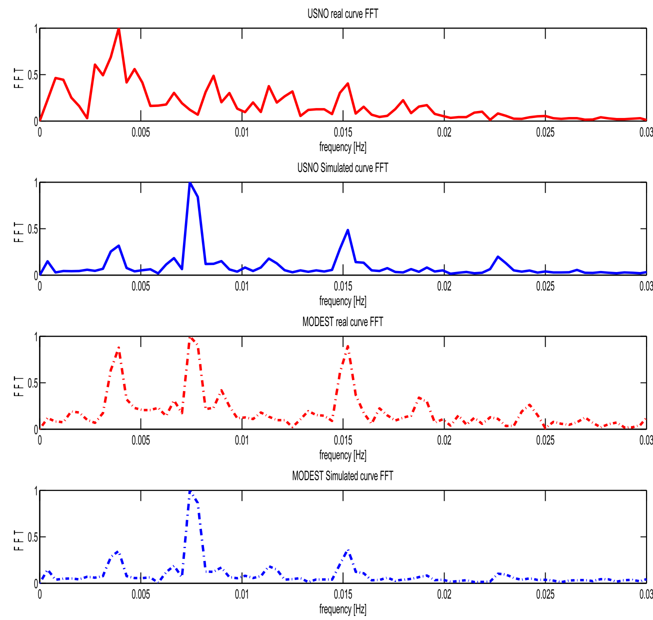

Fig. 9. Fast Fourier transform comparison between real and simulated lightcurve from both sites.

TABLE III
Fast Fourier Transform Analysis of Real and Simulated Lightcurves

USNO real curve FFT	USNO simulated curve FFT	Modest measured curve FFT	Modest simulated curve FFT
0.0007813	0.0003906	0.0003906	0.0003906
0.002734	–	0.001563	–
0.003906	0.003906	0.003906	0.003906
0.004688	–	0.005859	–
0.006641	0.006641	0.006641	0.006641
0.008594	0.007422	0.007422	0.007422
0.01133	0.01133	0.008984	0.01133
0.0125	–	0.01094	–
0.01406	–	0.01328	–
0.01523	0.01523	0.01523	0.01523

TABLE IV
Reconstructed Attitude Motion Quaternion and Angular Rate Components

$q1$	−0.0090
$q2$	0.8806
$q3$	−0.2531
$q4$ (scalar)	−0.4006
p_0 [rad/s]	0.00028
q_0 [rad/s]	0.01454
r_0 [rad/s]	0.01876

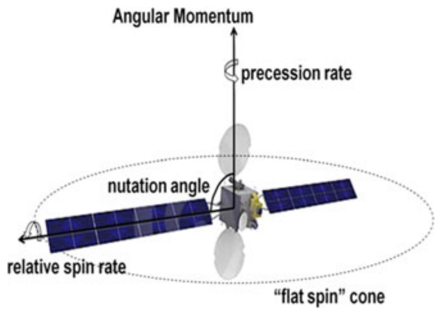

Fig. 10. Schematic representation of the GSAT3 reconstructed free-body motion according to the obtained measurements.

TABLE V
Dynamical Interpretation of the GSAT-3 Reconstructed "Free-Body" Attitude Motion

Initial parameter	Value
Nutation angle	89.6°
Precession rate	1.36°/s (0.23 rpm)
Relative spin rate	8e–3°/s (1.3e–3 rpm)
Angular momentum unit Vector components in J2000	−0.69 0.18 –0.70
Angular momentum Right Ascension (RA)	166°
Angular momentum declination (DEC)	−44°

The amplitudes were normalized, since once again, they are related to virtual reality tool rendering and not to attitude motion.

B. GSAT3 Attitude Motion

The results of attitude motion reconstruction for GSAT3 are summarized in Table IV, in which the components of the satellite body attitude quaternion with respect to the J2000 inertial reference frame and the angular velocity components in the body reference frame are listed.

The solar array rotation axis in the satellite reference frame can be considered as a symmetry axis for the spacecraft ellipsoid of inertia. Based on this simplifying assumption and in the absence of external torques, the resulting rigid body motion is a coning motion of the maximum axis of inertia about the inertially fixed angular momentum vector [25].

Because of long term energy dissipation, the satellite ends up in a typical "flat spin" motion. The initial conditions in Table IV correspond to a coning motion schematically depicted in Fig. 10, with the data indicated in Table V. The relative spin rate is very low compared to the precession rate, hence the motion can be assimilated to a pure spin about the angular momentum vector. The angular momentum declination indicates that the satellite spin axis is quite far from the orbit normal.

V. CONCLUSION

A novel approach to determine attitude motion of an orbiting object compatible with lightcurves achieved by optical observations was presented, based on virtual reality tools. The suitability of this method was demonstrated by application to a real measurement campaign of the satellite GSAT3 conducted from the Cerro Tololo and Flagstaff Observatories.

The method identified a set of rotation parameters of GSAT3 compatible with the measured lightcurves, except for few spikes, possibly due to real satellite elements and components not included in the virtual reality model. The performance and obtainable accuracy of the proposed approach are strictly related to the accurate knowledge of the shape, material, and reflecting properties of the target. Nevertheless, based on the results shown in this paper, it seems that even raw models of complex bodies can permit to achieve results compatible with measurements. The method is based on the reflectance irregularities of the satellite surfaces; hence, it fails for spherically symmetrical objects. Therefore, before attempting to use the algorithm with real measurements, it is extremely important to evaluate the degree of success of the attitude determination process for a particular spacecraft by using simulated noisy measurements.

This method can be used with measurements in different ranges of the reflectance spectra and can be adopted both for synchronized and not synchronized observations, and from monostatic or multistatic observations.

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